

Mini-project report.pdf

by Yuki Wu

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Coursework Coversheet

Name	Xiaoqi (Yuki) Wu
Date	20/03/2024
Module	Behavioural Ecology
Coursework item	Mini-project

Self-reflection is a non-assessed part of the coursework submission to aid your learning

1. Have you used previous feedback/self-reflection to produce this coursework? If so, how? If not, why?

Yes, I utilized my previous feedback to improve my time management by creating a well-organized schedule for completing this project.

2. What do you feel went well? (e.g., development of specific skills, sense of achievement, part of the assignment you are most proud of)

I did well with time management. I made a point to complete one section every day, allowing ample time for essay revisions.

I found the peer review session highly beneficial. I received several valuable suggestions and gained insights from reviewing my peer's essay. *great!*

3. What do you feel did not go well? (e.g., learning outcomes that you found most difficult to meet, aspects of the assignment that were challenging, time-management)

What I found challenging was finding related papers to discuss, as there were no papers that had conducted experiments to investigate the relationship between exposure time and cricket aggressive behaviours. Therefore, I focused more on discussing the limitations of our experiment.

Additionally, I discovered that it is quite difficult to make an experiment perfect, given the multitude of variables that need to be considered and controlled.

I also encountered difficulty in aligning the format of my essay with the sample paper.

4. What did you hope to learn from doing this coursework? Did you achieve that?

This is the first time we have designed our experiment independently. I anticipated learning about the considerations necessary for experiment design, which I have successfully achieved.

Exploring Variations in Cricket Aggression with Increasing Exposure Time to Higher Temperatures

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INTRODUCTION

The impact of global warming becomes more significant. It not only affects environments but also profoundly influences life on Earth. Among animals influenced by the increasing global temperature, research suggests that higher temperatures can have varying effects on insect behaviours. An increase in the frequency or duration of high temperatures resulting from climate change is likely to result in an immediate reduction in foraging efficiency (Andrew et al., 2013). It has been found that crickets raised at higher temperatures are more stable in their exploratory behaviours and produce faster mating calls with a higher peak frequency compared to those raised at lower temperatures (Niemelä et al., 2019, Singh, Prathibha & Jain, 2020). In a study conducted by Nguyen and Stahlschmidt (2019), it was shown that crickets tend to be more aggressive, and their fight will be longer under cooler morning conditions.

Based on this study, we expected that at higher temperatures (30°C), cricket aggressive levels would decrease compared to room temperature (23°C). Since no research has been conducted on the duration of exposure to warmer temperatures and its impact on crickets, our study focuses on cricket aggression and how increased exposure time to higher temperatures affects their aggressive behaviour using Mediterranean field crickets (*Gryllus bimaculatus*). As the change in temperature might take time to show a change in behaviours, we hypothesized that the longer the cricket is at a warmer temperature, the lower their aggressive levels. For experiment, we considered the aggressive levels through three different dimensions: the number of encounters, the total duration of encounters, and the highest level of aggression.

By studying how Mediterranean field crickets react to temperature changes, we hope to understand more about the relationship between environmental conditions and animal behaviour, particularly in the context of climate change and its implications for ecological systems.

METHODS

Experiment Procedure and Data Collection

The Mediterranean field crickets were ordered from livefoods.co.uk and maintained at 12:12hr light:dark cycle at 23°C in a CT/lab room. Four replicates were conducted, each involving the selection of eight male crickets raised at room-temperature conditions (approximately 23°C). The experiment

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MP-Introduction

was conducted in the morning, as their circadian rhythm influences crickets' activity (Nguyen & Stahlschmidt, 2019). The crickets were placed into isolated Falcon tubes, with four individuals from one box and the remaining four from another. Subsequently, we paired them up into four pairs based on their size, ensuring that each pair originated from different boxes. Minimizing size differences was crucial,  4 **Increased disparities in size may escalate the winning chance for the cricket with larger weaponry and hence shorter encounter duration (Judge & Bonanno, 2008).**

One pair was served as a control group and not subjected to elevated temperatures. Two males were positioned on opposite sides of an arena with a divider in the middle, allowing them to acclimate for five minutes. After the acclimation period, the divider was removed, and timing commenced immediately. Over the following 10 minutes, two observers recorded the number of encounters, the total duration of encounters, and the highest level of aggression. The levels of encounters are categorized into six: mutual avoidance (level 0), pre-established dominance (level 1), antennal fencing (level 2), unilateral mandible spreading (level 3), bilateral mandible spreading (level 4), mandible engagement (level 5), and grappling (level 6; Alexander, 1961).

To ensure the arena was free of odors, it was prepared using alcohol wipes, and a five-minute wait was observed to ensure complete evaporation of the alcohol. Subsequently, the remaining three pairs of crickets were placed on a pre-heated heat mat set to 30°C, and timing began immediately. After 20 minutes, one pair of crickets was introduced into the arena, followed by a five-minute acclimation period and 10 minutes of observation for aggressive behaviours. This process was repeated for the other two pairs at 40 minutes and 60 minutes, respectively.

Statistical Analysis

R version 3.3.0+ was used for all statistical analyses. Data from the cricket experiment were examined to assess the impact of exposure time at higher temperatures on aggressive behaviour. To ensure reliability, two observers were recording the data during the experiment. Agreement between observers was assessed using Kendall's coefficient of concordance test, showing strong agreement (coefficient = 0.995) for the highest aggression level, and confirming the reliability of our data. As our data is inter-reliable, we only used data from observer 1 for the following analysis.

MP-IOR

Welch Two Sample t-tests were conducted to prove whether an increase in temperature has a significant impact on the number of encounters, total duration of aggression, and highest level of aggression. For the response variables "number of encounters" and "highest level of aggression," we chose to use Poisson regression models and linear regression for "total duration of aggression". Linear regression is appropriate when the response variable is continuous and normally distributed, which is often the case for total duration data.

Generalized linear models (GLMs) were fitted with appropriate distributions and link functions for each response variable. For the Poisson regression models, we specified a Poisson distribution with a log link function to model the count data. For the linear regression model, we used a Gaussian

distribution with an identity link function to model the continuous outcome. Model analysis using the Q-Q plot checked that residuals are normally distributed for all three models.

Explanatory variables included in the models were exposure time (minutes), temperature ($^{\circ}\text{C}$), and any relevant covariates, such as the size difference between crickets. We initially intended for temperature and size differences to serve as control variables. However, due to the unstable heating of the heat mat and the challenge of ensuring uniform cricket sizes, we decided to also consider temperature and size differences as potential factors, as they could influence cricket behaviour.

MP-Data Analysis

Ethical Note

All study animals were ethically sourced and reared specifically for study purposes. No regulatory approval was required due to minimal risk. In our experiment, we allowed male crickets to fight each other. This is a common behaviour among crickets, allowing them to compete for resources and mating chances. Throughout the experiment, the welfare of the crickets was ensured through suitable housing conditions, nutrition, and stress minimization.

MP-Ethical Statement

RESULTS

Welch Two Sample t-tests showed a significant difference in both the number of encounters ($p = 0.040$) and total duration of encounters ($p = 0.013$) between control groups and groups exposed to higher temperatures. However, no significant difference is shown in the highest level of aggression ($p = 0.438$).

Stats-vague

The Poisson regression plot illustrates a gradual decrease in the number of encounters with increasing exposure time (Fig. 1a). The regression coefficient for exposure time was estimated at -0.010 , indicating a negative association between exposure time and encounter frequency (Table 1). However, the analysis of the number of encounters among male crickets by the Poisson regression model revealed no significant relationship with exposure time under varying temperatures ($p = 0.269$; Table 1). Temperature (estimate = -0.090 , $p = 0.029$) and size difference (estimate = -3.065 , $p = 0.023$) showed significant effects on encounter frequency (Table 1).

Analysis of the total duration of aggressive encounters in male crickets using a linear regression model indicated a negative trend with exposure time, although the relationship was not statistically significant (estimate = -0.421 , $p = 0.105$; Table 1). The negative trend is also shown in Figure 1b. In addition, the effects of temperature ($p = 0.189$) and size difference ($p = 0.348$) showed no statistical significance as well (Table 1).

Investigation into the highest level of aggression displayed by male crickets through a Poisson regression model revealed no significant relationship with exposure time (estimate = -0.0116 , $p = 0.240$), temperature ($p = 0.832$), or size difference ($p = 0.519$; Table 1). Figure 1c illustrates the Poisson regression plot, indicating a slight decrease in the highest level of aggression with longer exposure times.

These results reveal that an increase in temperature has a significant impact on the number of

encounters and total duration of encounters, but no significant effect was observed on the highest level of aggression. Exposure time exhibits a minimal impact on encounter frequency and aggression levels. Moreover, the relationship between exposure time and total duration is not statistically significant, suggesting that other factors may play a more substantial role in determining the duration of aggressive encounters. Although the relationship was not statistically significant, a trend towards reduced aggression levels with longer exposure times was observed.

Table 1 Summary of estimate values and p-values for exposure time, temperature, and size difference as explanatory variables across three regression models. The table depicts estimates and p-values from Poisson regression models for encounters and aggression, and a linear regression model for duration. Negative estimates signify negative correlations, with only two p-values <0.05, indicating significant associations between encounters and temperature, and size difference.

Characteristic	Number of encounters		Total duration		Highest level of aggression	
	Estimate	p-value	Estimate	p-value	Estimate	p-value
Exposure time	-0.010	0.269	-0.421	0.105	-0.012	0.240
Temperature	-0.090	0.029 *	-1.624	0.189	-0.009	0.832
Size difference	-3.065	0.023 *	-43.150	0.348	-1.064	0.519
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1						

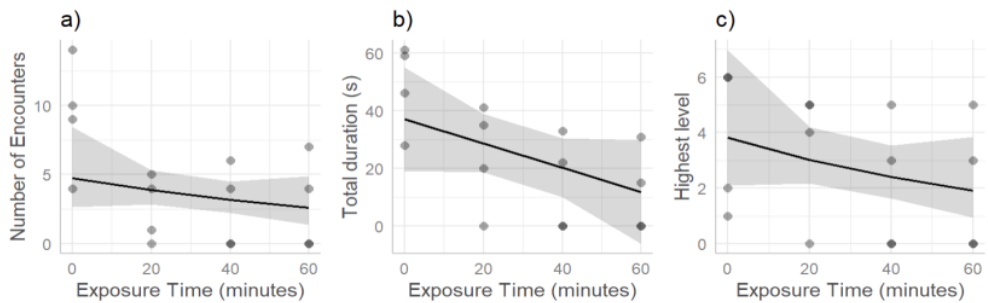


Fig. 1 Effects of exposure time on encounters, duration, and aggression in male crickets. The lines show the expected trend of each explanatory variable as exposure time varies, based on the fitted model. The shaded regions represent the 95% confidence interval around the predicted relationship. **a)** Relationship between exposure time and number of encounters observed in male crickets. **b)** Impact of exposure time on total duration of encounters in male crickets. **c)** Influence of exposure time on highest level of aggression displayed by male crickets, analysed by Poisson regression model.

DISCUSSION

Although it is commonly agreed that a lower temperature inhibits activity and a higher temperature usually stimulates the animal (Mellanby & Gardiner, 1997), we have demonstrated for Mediterranean field crickets that warmer temperatures indeed reduce the aggressiveness of their behaviours in terms of both the number of encounters and the total duration of encounters, corroborating the findings of Nguyen and Stahlschmidt (2019).

We expected that exposure time at higher temperatures might have significant effects on cricket aggression. However, there is no significant relationship between exposure time and cricket aggression across all three dimensions of measurement. This might be due to exposure intervals being too large, as changes in environmental temperature almost immediately affect crickets' body temperature and their behaviours. Therefore, shorter exposure intervals, such as 5 minutes, could be used for further investigation.

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Furthermore, sample size matters significantly. When considering data from both sets of observers instead of just one, there is a significant impact on the total duration of encounters (estimate = -0.007, $p < 0.05$). This might be because with a larger sample size, the standard error decreases. This indicates that with more samples taken, it becomes possible to demonstrate a significant impact on the number of encounters and the highest level of duration.

Another limitation is that the real data does not adhere to the assumed conditions of the GLM model. The GLM model assumes that cricket aggression behaviour is independent at every unit of time as exposure time to warmer temperatures progresses. However, this assumption is not hold true because cricket aggression depends on their previous encounter status.

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Apart from that, if temperature and cricket size were perfectly controlled, different results might have been shown. Additionally, there are other variables that have not been considered, such as antioxidant defence abilities. Under warmer conditions, crickets might have higher metabolic rates, hence producing more reactive oxygen species. Crickets have varying antioxidant defence abilities, which impact their acclimatization to increased temperatures (Lorenz, Fischer & Beaulieu, 2020). Therefore, the differences in antioxidant defence abilities among cricket individuals might also affect the degree of change in their aggressive behaviours.

In conclusion, a 7°C increase in temperature decreases cricket aggressive activity, while exposure time under warmer temperatures shows no significant relationship with aggressive behaviours. These findings suggest that a rise in temperature, possibly due to global warming, could reduce cricket aggressive activity immediately. Further investigations could explore smaller exposure time intervals, larger sample sizes, and carefully controlled cricket size and temperature.

7

2026 words

MP-Discussion

REFERENCE

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APPENDIX

expo.time.min	size1	size2	size.diff	temp	no.enc1	no.enc2	total.duration.s1	total.duration.s2	highest.level1	highest.level2
0	6.1	6.28	0.18	24	14	14	46	46	6	6
20	6.88	6.95	0.07	29.4	4	3	20	29	5	5
40	6.71	7.01	0.3	33	0	0	0	0	0	0
60	7.1	7.29	0.19	31	4	4	15	17	3	3
0	6.39	6.62	0.23	23	4	4	28	20	1	1
20	6.71	7.01	0.3	29.4	1	1	35	35	4	5
40	6.78	6.8	0.02	34.6	0	0	0	0	0	0
60	6.58	6.72	0.14	31	0	0	0	0	0	0
0	6.39	6.44	0.05	23	10	10	59	67	2	1
20	7.26	7.45	0.19	36	0	0	0	0	0	0
40	5.95	6.12	0.17	34.5	6	8	33	36	3	2
60	6.42	6.54	0.12	33	0	0	0	0	0	0
0	6.68	6.94	0.26	23	9	10	61	84	6	6
20	6.34	6.38	0.04	36	5	5	41	39	5	5
40	6.14	6.33	0.19	34.5	4	4	22	25	5	5
60	6.14	6.14	0	33	7	7	31	35	5	5

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GRADEMARK REPORT

FINAL GRADE

GENERAL COMMENTS

72/100

PAGE 1

Text Comment. great!

PAGE 2



Comment 1

what is the impact of this on fitness?



Comment 2

justified hypothesis- well done



MP-Introduction

Introduction should set the scene for the study giving the scientific background to the study e.g. the theory/observations, key concepts (eg impacts of size in aggression) followed by why your study is important (eg the gaps in the study area), the aims of the study and any relevant predictions/hypotheses

Additional Comment

You need to make the link to how climate change affects animal behaviour and affects ecosystem functioning- can you give an example to justify the importance of studying it?



Comment 3

n= 4 is very low.

PAGE 3



MP-Cricket apparatus

It would be useful to mention the size of arenas as this could impact the behaviour. Also perhaps think about using images here either as diagrams or photos of the apparatus-it will help the reader visual your expt and could save words.



Comment 4

is it always shorter in duration? Good to link to body size and this could have been mentioned in the Intro also.



MP-IOR | Criteria met

It is good to do in inter-observer reliability test to ensure your behaviour scoring and therefore your data are robust. You should report the methods and the results in the Methods section of your paper; usually after you have detailed the experimental protocol. State what variable you observed, IOR test you performed and the resulting metric

PAGE 4



MP-Data Analysis | Criteria met

Studies often have a data analysis section at the end of the methods. A reason for doing this is so that the results section isn't too dense with very detailed description of stats procedures and hard to follow.

Additional Comment

excellent section- very clear



MP-Ethical Statement

is required and is good to show you have thought about the welfare of the organisms- eg the crickets were given access to shelter, and *ad libitum* water and food in their home crates. It is always a good idea to refer to the ASAB guidelines (eg ASAB, 2023) and put them in the ref. list under ASAB



Stats-vague

You have not given all the relevant information here. You have either not said how the stats have been performed or just given p values without any other information (such as test statistic, df/N) or both

PAGE 5

Text Comment. excellent table



MP-Results

The Results section should guide the reader through the analysis that was done and why in a clear easy to follow format- where the logic is clear. It usually has plots or Tables to show the basic descriptive stats and therefore give an initial look see at the differences between treatments. Graphics are generally easier for the reader to get a feel for the data. Also do not forget, they also bring the biology into the Results section eg how many encounters in the presence and in the absence of a female so the Results are not just a series of stats tests. ALWAYS start with an introductory paragraph - do not show a graphic straight under the Results title. If you are reporting GLMs- think about putting the relevant details in a Table ie the explan. variables with their coeff, +/- SE, z/t value and p-value. It will save a lot of words and will read more easily; but don't forget to interpret it in the main text.

Additional Comment

generally well reported and good graphics.

PAGE 6



Comment 5

yes!



Comment 6

bring in winner loser effects also



Comment 7

what would the impact of this be?



MP-Discussion

The Discussion could start by simply stating your hypotheses and main findings to remind the reader, especially if you have a complicated Results section. Then you should set YOUR results in context of what is know already known from previous research. It should also make suggestions for future work to distinguish between various possible explanations. There is therefore an expectation that relevant references will be included in the Discussion to allow you to set your results in context. You should **briefly** discuss limitations, to acknowledge they exist but not spend a significant amount of time over them with a relatively low word count

PAGE 7

GRADING FORM: LIFESCIGRADINGFORMMINORV1

YUKI WU

CRITERIA MET

How does this work meet the criteria for the grade it has been given?



This is an interesting idea for a project looking at heat exposure time and aggression. You raised interesting hypotheses and the methods were clear, but the level of replication was very low for each of the timed treatments. Perhaps it would have been better to have fewer times and more replication. The report is well written and so the methods and stats analysis section are very well written- Table 1 is excellent

AREAS TO IMPROVE

How could this work – and similar work in future – be improved, to gain a higher mark?



You do note sample size is an issue and so needed more consideration at the design stage. It would have been good to also speak more about size/RHP more- for example you would expect evenly matched males to show higher levels of aggression, what impact would this have on your study. See further comments in the text.